Benchmark Specifications

Paper	*Time-to-reach Bounds for Verification of Dynamical Systems Using the Koopman Spectrum*
Content	The 16 benchmarks of Table II: the specification of each and the result of every tool
Code	https://github.com/Aalto-Nonlinear-Systems-and-Control/Time-to-Reach; every benchmark is the JSON file `configs/benchmarks/`*config ID*`.json`
Online report	https://aalto-nonlinear-systems-and-control.github.io/Time-to-Reach/
Results	The row of Table II: verdict (computation time in seconds)
Verdicts	Rea: reachable; PRea: periodically reachable; Unr: unreachable; Ukn: unknown; Inc: inconclusive; δ-Rea: δ-sat of dReach
No verdict	N/A: not applicable; S/E: set explosion; OOM: out of memory; T/O: time-out

1 Duffing's oscillator (Example 1)

1.1 System

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} x_2 \\ -0.5x_2 - x_1(x_1^2 - 1) \end{bmatrix}$$

1.2 Benchmark DUFF_REA_BOX

Config ID: `DUFF_FIN_REA_BOX`.

1.2.1 Specification

- Initial Set: $X_0 = [-0.8, -0.7] \times [1.3, 1.39]$
- Target Set: $X_F = [0.7, 0.8] \times [0.178, 0.3]$
- Time Horizon: $[0, 6]$
- Expected Result: Reachable

1.2.2 Results

KRTB		CORA	JuliaReach	dReach	hj_reachability	SOSTOOLS
Eigenpairs	Bounds					
4.81	Rea (0.01)	Ukn (9.19)	Ukn (1.27)	δ-Rea (446.86)	Rea (19.81)	Ukn (1.70)

KRTB reach-time bounds: $[5.396, 5.658]$. Simulation baseline: 63 of 2004 sampled trajectories reach the target, at the times $[5.608, 5.643]$.

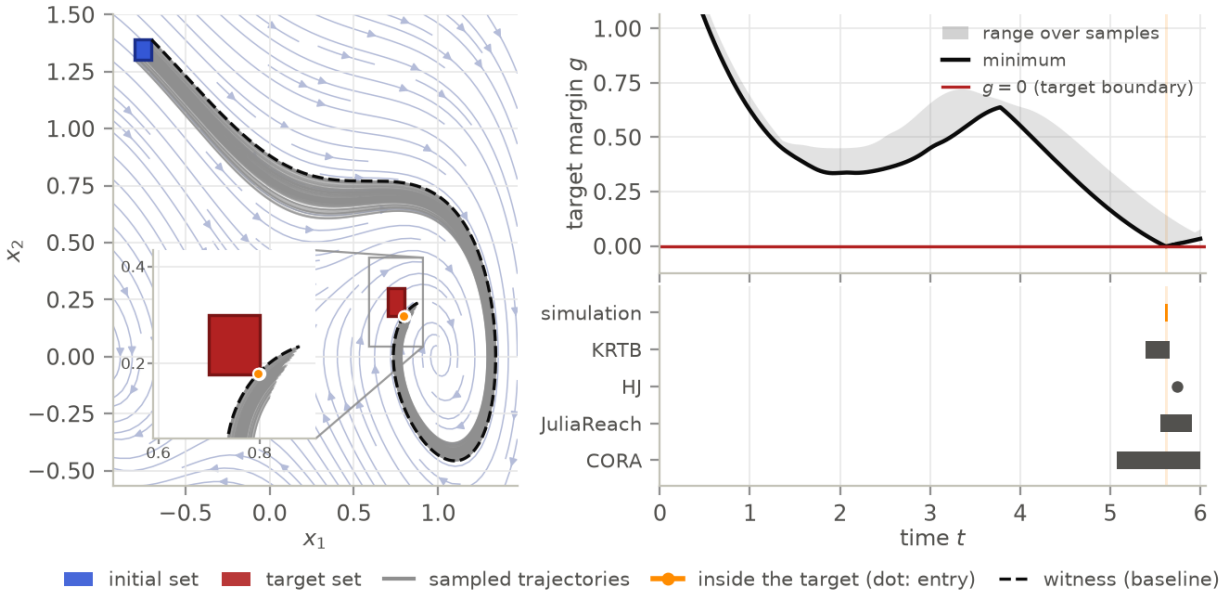
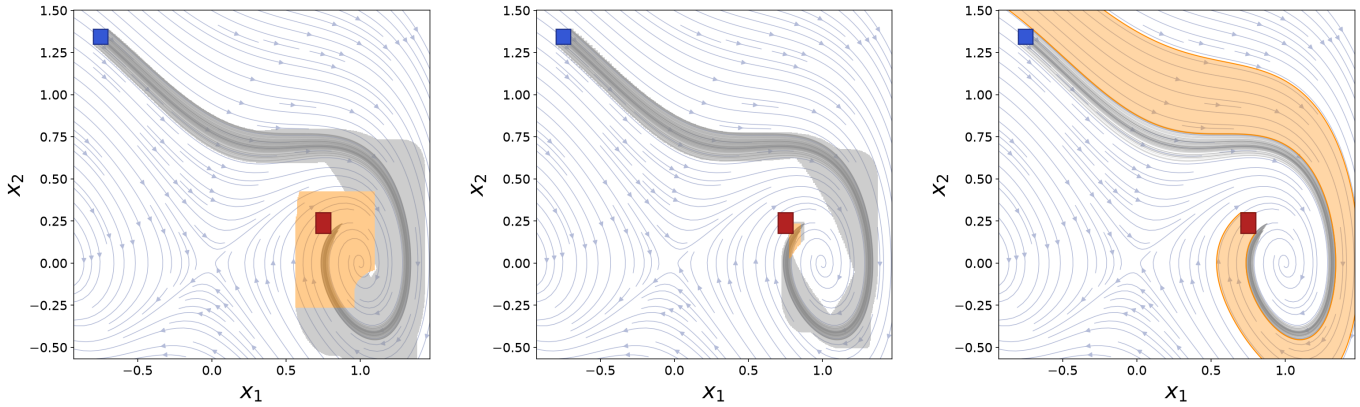


Figure 1: DUFF_REA_BOX: 84 sampled trajectories from X_0 (grey; orange inside X_F, with a dot at the entry), the witness of the simulation baseline (dashed), the target margin $g(x(t))$ ($g \leq 0$ inside X_F) and, on the same time axis, the simulated in-target times (orange band), the KRTB reach-time bounds, the first reach time estimated by hj_reachability (dot), the times of the JuliaReach flowpipe segments that may meet the target and the times of the CORA flowpipe segments that may meet the target.



(a) CORA: Ukn (9.19).

(b) JuliaReach: Ukn (1.27).

(c) hj_reachability: Rea (19.81).

Figure 2: DUFF_REA_BOX: what the tools computed, drawn in the phase plane with the sampled trajectories (grey). CORA and JuliaReach: the flowpipe, with the segments disjoint from the target in grey, those that may meet it in light orange and those inside it in orange; hj_reachability: the states from which the target is reached ($V \leq 0$, light orange).

1.3 Benchmark DUFF_REA_CVX_LS

Config ID: `DUFF_FIN_REA_BALL`.

1.3.1 Specification

- Initial Set: $X_0 = \{x \in \mathbb{R}^2 \,|\, (x_1 + 0.75)^2 + (x_2 - 1.78)^2 - 0.01 \leq 0\}$
- Target Set: $X_F = \{x \in \mathbb{R}^2 \,|\, (x_1 - 1.3)^2 + (x_2 + 0.8)^2 - 0.01 \leq 0\}$
- Time Horizon: $[0, 6]$
- Expected Result: Reachable

1.3.2 Results

KRTB		CORA	JuliaReach	dReach	hj_reachability	SOSTOOLS
Eigenpairs	Bounds					
4.90	Rea (0.01)	N/A	N/A	δ-Rea (14.31)	Rea (21.61)	Ukn (1.43)

KRTB reach-time bounds: $[1.530, 2.614]$. Simulation baseline: 79 of 2000 sampled trajectories reach the target, at the times $[2.562, 2.613]$.

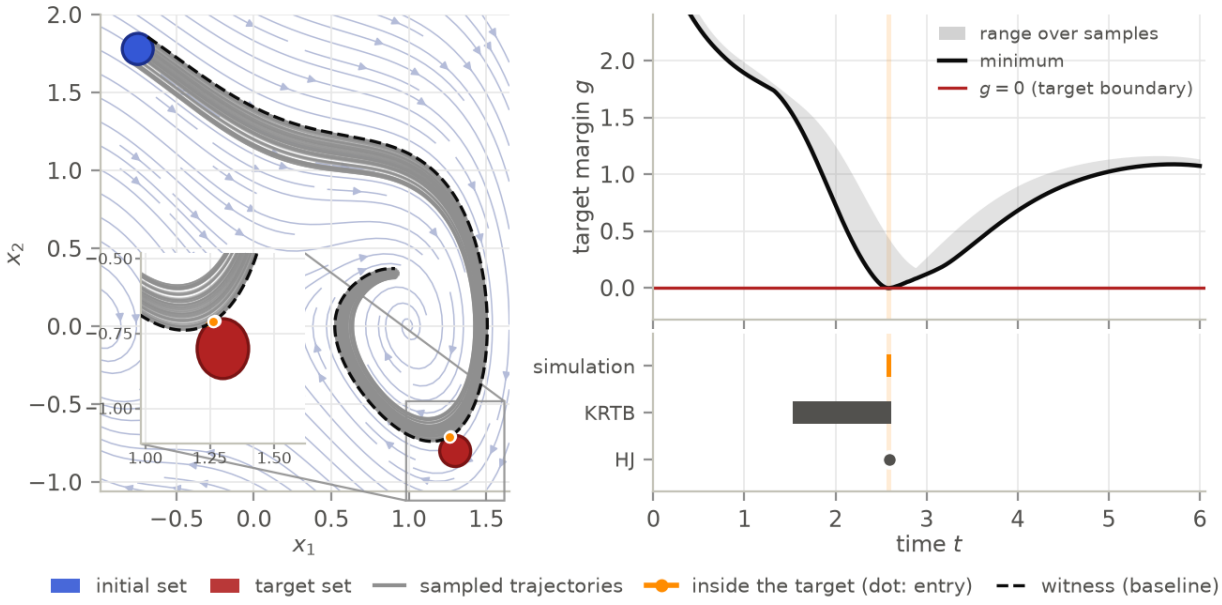
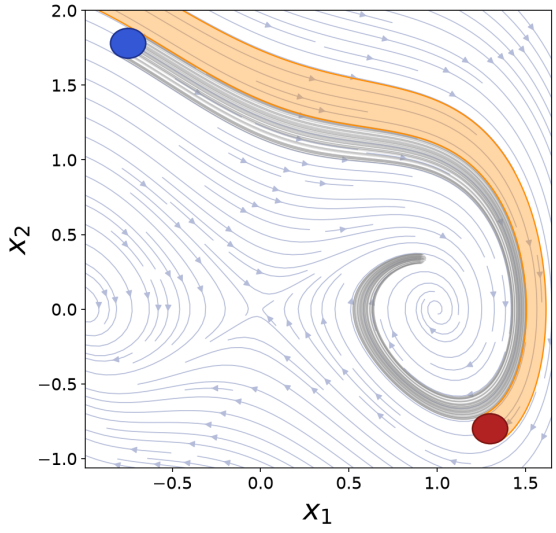


Figure 3: DUFF_REA_CVX_LS: 80 sampled trajectories from X_0 (grey; orange inside X_F, with a dot at the entry), the witness of the simulation baseline (dashed), the target margin $g(x(t))$ ($g \leq 0$ inside X_F) and, on the same time axis, the simulated in-target times (orange band), the KRTB reach-time bounds and the first reach time estimated by hj_reachability (dot).



(a) hj_reachability: Rea (21.61).

Figure 4: DUFF_REA_CVX_LS: what the tools computed, drawn in the phase plane with the sampled trajectories (grey). Hj_reachability: the states from which the target is reached ($V \leq 0$, light orange).

1.4 Benchmark DUFF_REA_NCVX_LS

Config ID: `DUFF_FIN_REA_NONCONVEX_LEVEL_SET`.

1.4.1 Specification

- Initial Set: $X_0 = \{x \in \mathbb{R}^2 \mid -\left(1 - \frac{x_1+0.75}{3} + \left(\frac{x_2-1.85}{0.05}\right)^5 + 2\left(\frac{x_1+0.75}{0.05}\right)^3\right) \exp\left(-\left[\left(\frac{x_1+0.75}{0.05}\right)^2 + \left(\frac{x_2-1.85}{0.05}\right)^2\right]\right) + 0.01 \leq 0\}$
- Target Set: $X_F = \{x \in \mathbb{R}^2 \mid -\left(1 - \frac{x_1-0.65}{3} + 3\left(\frac{x_2-0.25}{0.1}\right)^5 + 4\left(\frac{x_1-0.65}{0.1}\right)^3\right) \exp\left(-\left[\left(\frac{x_1-0.65}{0.1}\right)^2 + \left(\frac{x_2-0.25}{0.1}\right)^2\right]\right) + 0.9 \leq 0\}$
- Time Horizon: $[0, 6]$
- Expected Result: Reachable

1.4.2 Results

KRTB		CORA	JuliaReach	dReach	hj_reachability	SOSTOOLS
Eigenpairs	Bounds					
4.78	Rea (0.01)	N/A	N/A	δ-Rea (1589.39)	Rea (16.19)	N/A[a]

[a] Not directly applicable due to trigonometric, exponential or fractional polynomial functions.

KRTB reach-time bounds: $[5.048, 5.831]$. Simulation baseline: 2000 of 2000 sampled trajectories reach the target, at the times $[5.051, 5.808]$.

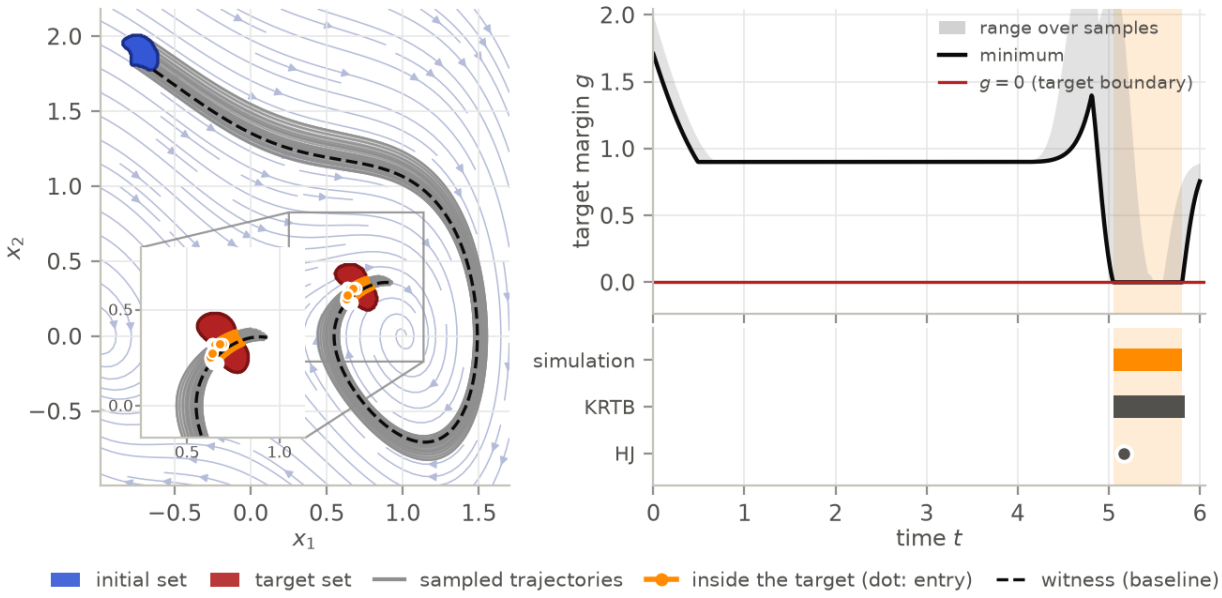
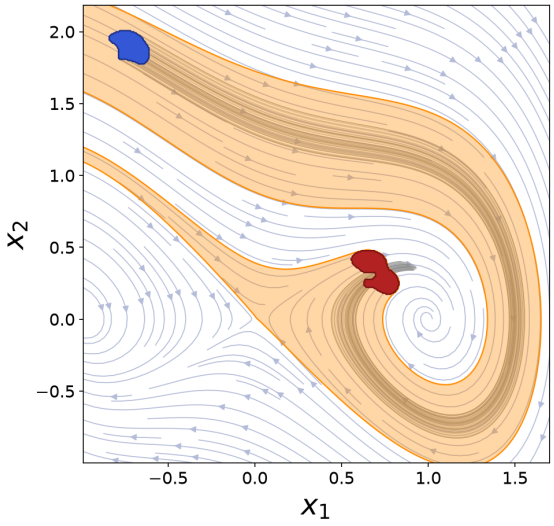


Figure 5: DUFF_REA_NCVX_LS: 80 sampled trajectories from X_0 (grey; orange inside X_F, with a dot at the entry), the witness of the simulation baseline (dashed), the target margin $g(x(t))$ ($g \leq 0$ inside X_F) and, on the same time axis, the simulated in-target times (orange band), the KRTB reach-time bounds and the first reach time estimated by hj_reachability (dot).



(a) hj_reachability: Rea (16.19).

Figure 6: DUFF_REA_NCVX_LS: what the tools computed, drawn in the phase plane with the sampled trajectories (grey). Hj_reachability: the states from which the target is reached ($V \leq 0$, light orange).

1.5 Benchmark DUFF_REA_DSJT_BOX

Config ID: `DUFF_FIN_REA_DISJOINT_BOX`.

1.5.1 Specification

- Initial Set: $X_0 = [-0.8, -0.7] \times [1.8, 1.9]$
- Target Set: $X_F = [0.6, 0.7] \times [0.2, 0.3] \cup [1.0, 1.1] \times [1.0, 1.1]$
- Time Horizon: $[0, 6]$
- Expected Result: Reachable

1.5.2 Results

KRTB		CORA	JuliaReach	dReach	hj_reachability	SOSTOOLS
Eigenpairs	Bounds					
7.37	Rea (0.01)	N/A	N/A	δ-Rea (49.06)	Rea (17.60)	Ukn (1.56)

KRTB reach-time bounds: $[0.692, 1.449] \cup [4.960, 5.963]$. Simulation baseline: 2004 of 2004 sampled trajectories reach the target, at the times $[1.246, 1.439] \cup [4.976, 5.421]$.

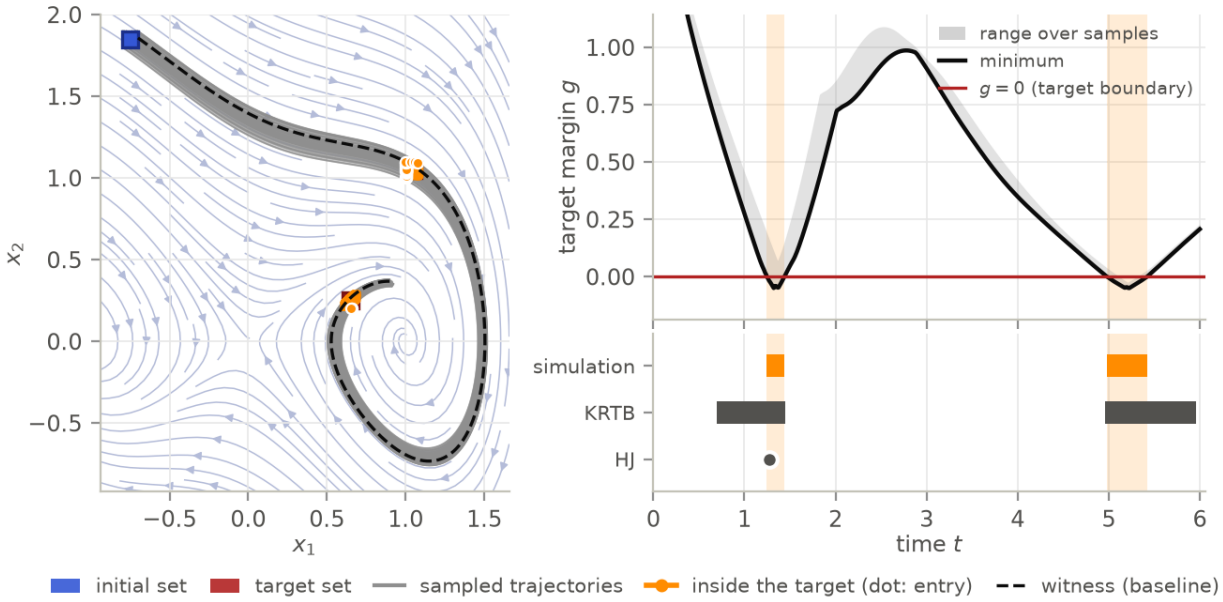
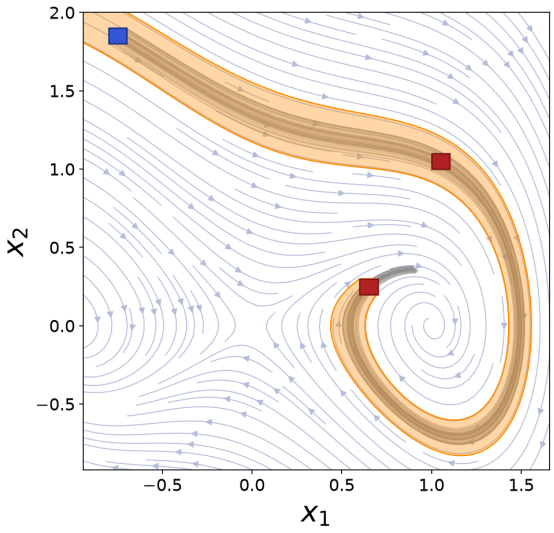


Figure 7: DUFF_REA_DSJT_BOX: 84 sampled trajectories from X_0 (grey; orange inside X_F, with a dot at the entry), the witness of the simulation baseline (dashed), the target margin $g(x(t))$ ($g \leq 0$ inside X_F) and, on the same time axis, the simulated in-target times (orange band), the KRTB reach-time bounds and the first reach time estimated by hj_reachability (dot).



(a) hj_reachability: Rea (17.60).

Figure 8: DUFF_REA_DSJT_BOX: what the tools computed, drawn in the phase plane with the sampled trajectories (grey). Hj_reachability: the states from which the target is reached ($V \leq 0$, light orange).

1.6 Benchmark DUFF_UNR_BOX

Config ID: `DUFF_FIN_UNR_BOX`.

1.6.1 Specification

- Initial Set: $X_0 = [-0.8, -0.7] \times [1.3, 1.35]$
- Target Set: $X_F = [0.7, 0.8] \times [0.178, 0.3]$
- Time Horizon: $[0, 20]$
- Expected Result: Unreachable

1.6.2 Results

KRTB		CORA	JuliaReach	dReach	hj_reachability	SOSTOOLS
Eigenpairs	Bounds					
4.87	Inc (0.01)	Ukn (9.97)	Ukn (1.40)	T/O	Unr (65.02)	Unr (1.33)

KRTB reach-time bounds: $[5.474, 5.567]$. Simulation baseline: none of 2004 sampled trajectories reaches the target; the closest approach (distance) is 0.009319.

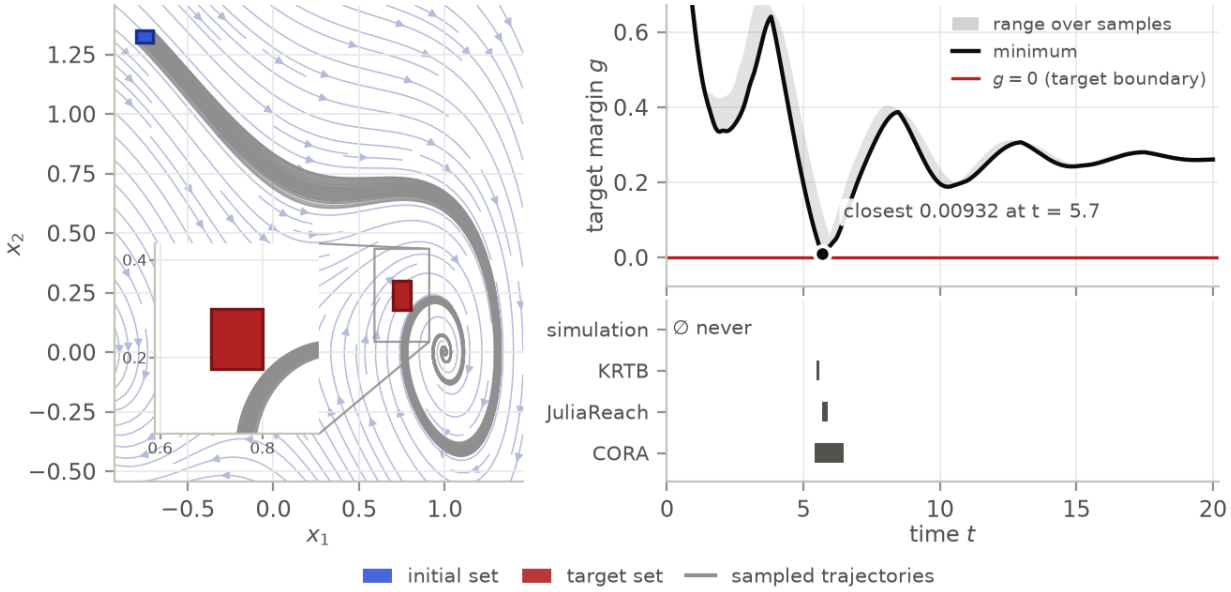
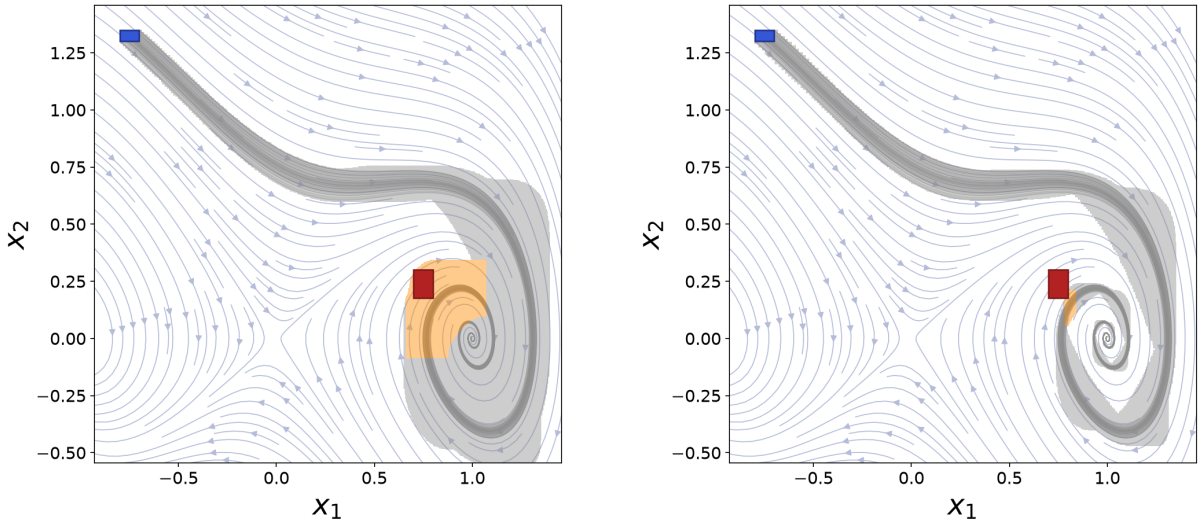
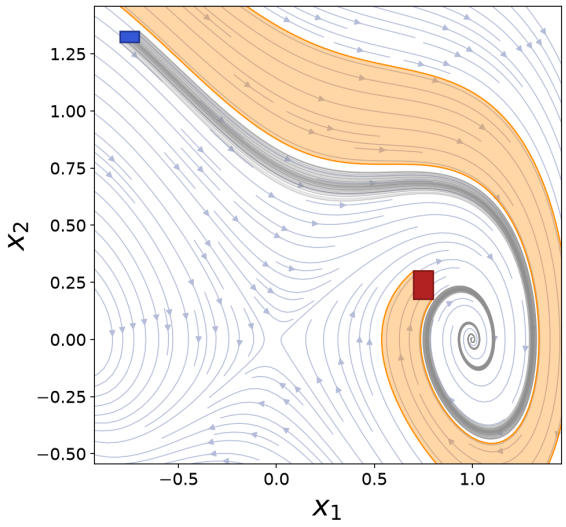


Figure 9: DUFF_UNR_BOX: 84 sampled trajectories from X_0 (grey), the target margin $g(x(t))$ ($g \leq 0$ inside X_F) and, on the same time axis, the simulated in-target times (none), the KRTB reach-time bounds, the times of the JuliaReach flowpipe segments that may meet the target and the times of the CORA flowpipe segments that may meet the target.

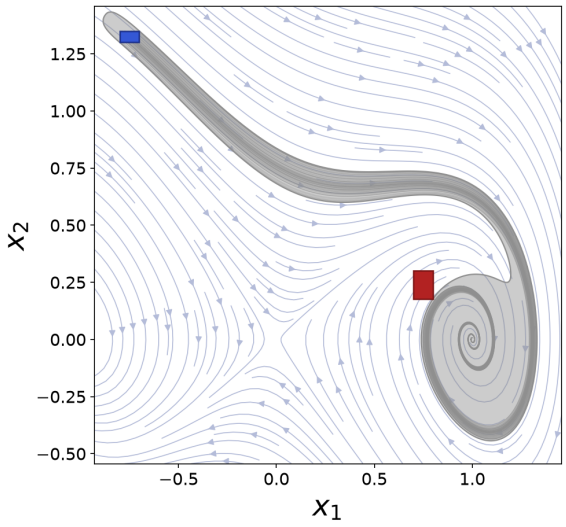


(a) CORA: Ukn (9.97).

(b) JuliaReach: Ukn (1.40).



(c) hj_reachability: Unr (65.02).



(d) SOSTOOLS: Unr (1.33).

Figure 10: DUFF_UNR_BOX: what the tools computed, drawn in the phase plane with the sampled trajectories (grey). CORA and JuliaReach: the flowpipe, with the segments disjoint from the target in grey, those that may meet it in light orange and those inside it in orange; hj_reachability: the states from which the target is reached ($V \leq 0$, light orange); SOSTOOLS: the region $B < 0$ of the barrier certificate (grey), which is certified never to reach the target.

1.7 Benchmark DUFF_UNR_CVX_LS

Config ID: `DUFF_FIN_UNR_BALL`.

1.7.1 Specification

- Initial Set: $X_0 = \{x \in \mathbb{R}^2 \,|\, (x_1 + 0.75)^2 + (x_2 - 1.75)^2 - 0.01 \leq 0\}$
- Target Set: $X_F = \{x \in \mathbb{R}^2 \,|\, (x_1 - 1.3)^2 + (x_2 + 0.8)^2 - 0.01 \leq 0\}$
- Time Horizon: $[0, 6]$
- Expected Result: Unreachable

1.7.2 Results

KRTB		CORA	JuliaReach	dReach	hj_reachability	SOSTOOLS
Eigenpairs	Bounds					
4.84	Inc (0.01)	N/A	N/A	Unr (1725.68)	Unr (22.03)	Unr (1.24)

KRTB reach-time bounds: $[1.526, 2.512]$. Simulation baseline: none of 2000 sampled trajectories reaches the target; the closest approach (distance) is 0.01671.

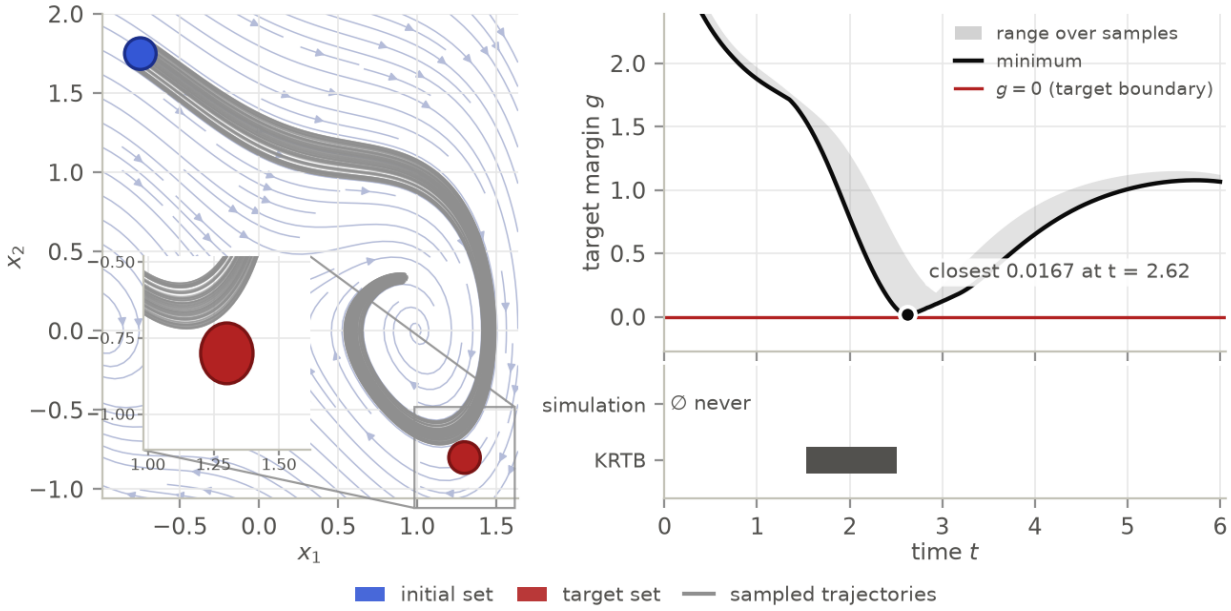
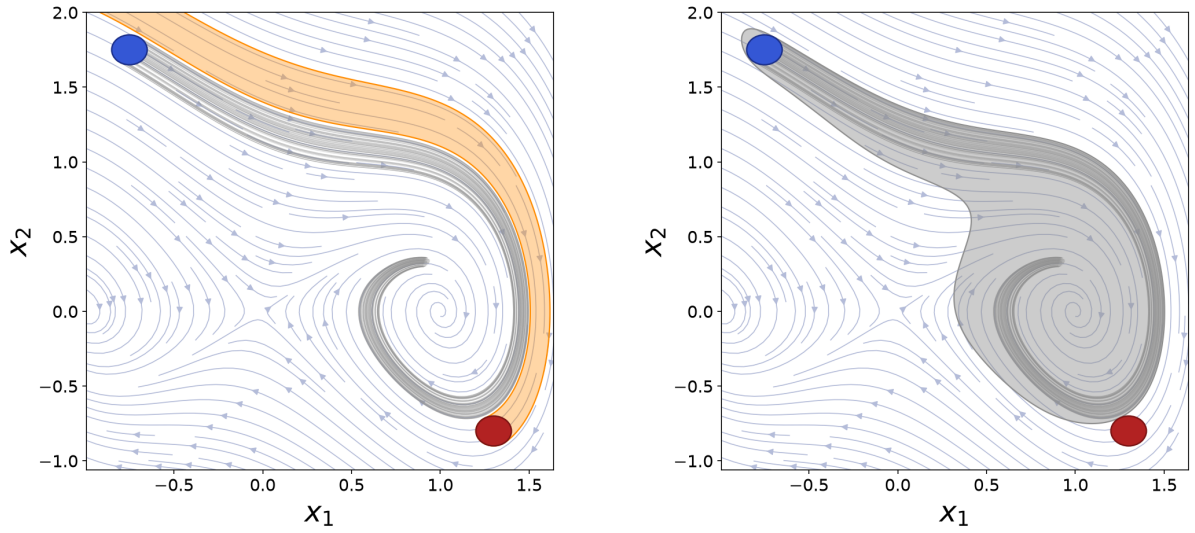


Figure 11: DUFF_UNR_CVX_LS: 80 sampled trajectories from X_0 (grey), the target margin $g(x(t))$ ($g \leq 0$ inside X_F) and, on the same time axis, the simulated in-target times (none) and the KRTB reach-time bounds.



(a) hj_reachability: Unr (22.03).

(b) SOSTOOLS: Unr (1.24).

Figure 12: DUFF_UNR_CVX_LS: what the tools computed, drawn in the phase plane with the sampled trajectories (grey). Hj_reachability: the states from which the target is reached ($V \leq 0$, light orange); SOSTOOLS: the region $B < 0$ of the barrier certificate (grey), which is certified never to reach the target.

2 Nonlinear System with Stable Limit Cycle

2.1 System

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} \frac{-x_1^3+4x_1^2x_2-x_1x_2^2+x_1+4x_2^3}{4(x_1^2+x_2^2)} \\ \frac{-4x_1^3-x_1^2x_2-4x_1x_2^2-x_2^3+x_2}{4(x_1^2+x_2^2)} \end{bmatrix}$$

2.2 Benchmark NL-SLC_REA_BOX

Config ID: `NL_SLC_FIN_REA_BOX`.

2.2.1 Specification

- Initial Set: $X_0 = [1.3, 1.4] \times [1.3, 1.4]$
- Target Set: $X_F = [\frac{\sqrt{3}}{2}, \frac{\sqrt{3}}{2} + 0.1] \times [0.5, 0.6]$
- Time Horizon: $[0, 15]$
- Expected Result: Reachable

2.2.2 Results

KRTB		CORA	JuliaReach	dReach	hj_reachability	SOSTOOLS
Eigenpairs	Bounds					
Analytical	PRea (0.01)	S/E	Ukn (2.51)	T/O	Rea (268.87)	N/A[a]

[a] Not directly applicable due to trigonometric, exponential or fractional polynomial functions.

KRTB reach-time bounds: $[6.426, 6.628] \cup [12.709, 12.911]$. Simulation baseline: 2004 of 2004 sampled trajectories reach the target, at the times $[6.428, 6.608] \cup [12.787, 12.866]$.

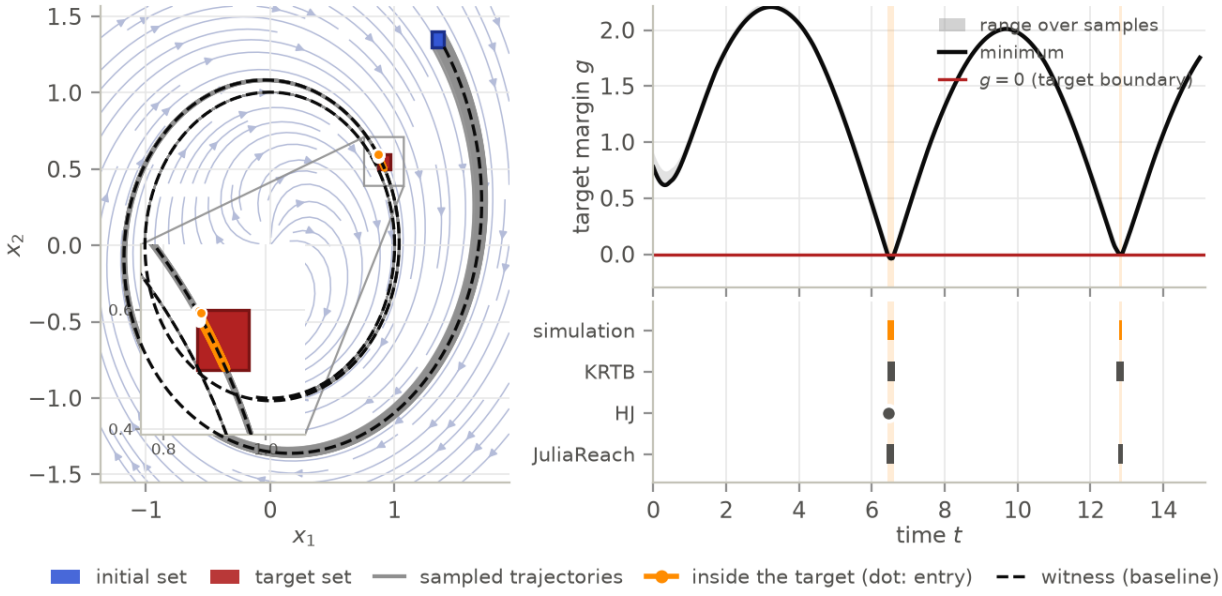
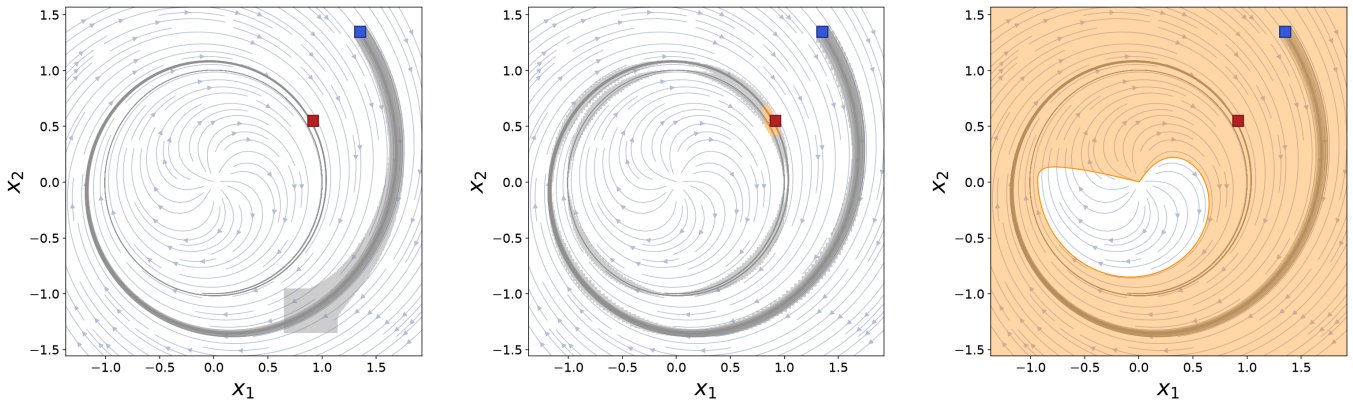


Figure 13: NL-SLC_REA_BOX: 84 sampled trajectories from X_0 (grey; orange inside X_F, with a dot at the entry), the witness of the simulation baseline (dashed), the target margin $g(x(t))$ ($g \leq 0$ inside X_F) and, on the same time axis, the simulated in-target times (orange band), the KRTB reach-time bounds, the first reach time estimated by hj_reachability (dot) and the times of the JuliaReach flowpipe segments that may meet the target.



(a) CORA: S/E. (b) JuliaReach: Ukn (2.51). (c) hj_reachability: Rea (268.87).

Figure 14: NL-SLC_REA_BOX: what the tools computed, drawn in the phase plane with the sampled trajectories (grey). CORA and JuliaReach: the flowpipe, with the segments disjoint from the target in grey, those that may meet it in light orange and those inside it in orange; hj_reachability: the states from which the target is reached ($V \leq 0$, light orange).

Note on periodic reachability: The figure below further illustrates the reachable sets for the benchmark NL-SLC_REA_BOX, computed using our method. Since the system has a stable limit cycle $\Gamma = \{x \,|\, \|x\| = 1\}$, trajectories that originated from the initial set X_0 may periodically pass through the target set X_F multiple times over some time horizon while converging to the limit cycle. To verify reachability under this challenging configuration, we consider the initial set $X_0 = [1.3, 1.4] \times [1.3, 1.4]$ and target set $X_F = [\frac{\sqrt{3}}{2}, \frac{\sqrt{3}}{2} + 0.1] \times [0.5, 0.6]$. Applying our proposed method, we obtain the estimated reach-time bounds as $I_0 = [6.426, 6.628]$ and $I_1 = I_0 + P = [12.709, 12.911]$ where period $P = 6.283$. The visualization in Fig. 15 validates our estimations. We note that, as X_F approaches Γ, $\|\underline{\psi_1}(X_F)\|$ and $\|\underline{\psi_2}(X_F)\|$ approach zero, since the analytical principal eigenfunctions for this system are $\psi_1(x) = (1 - x_1^2 - x_2^2)^2 e^{\arctan(x_2/x_1)j}$ and $\psi_2(x) = \psi_1^*(x)$. Therefore, the reach time interval obtained using eigenfunction modulus approaches infinity. On the other hand, the reach time estimated using eigenfunction phase would be $I_{phase} = I_0 + mP$ with $m \in \mathbb{Z}$, which can intersect $I_{mag} = (0, \infty)$ infinite number of times at periodic intervals, giving us periodic reach time estimates when X_F approaches the limit cycle Γ, as one would expect.

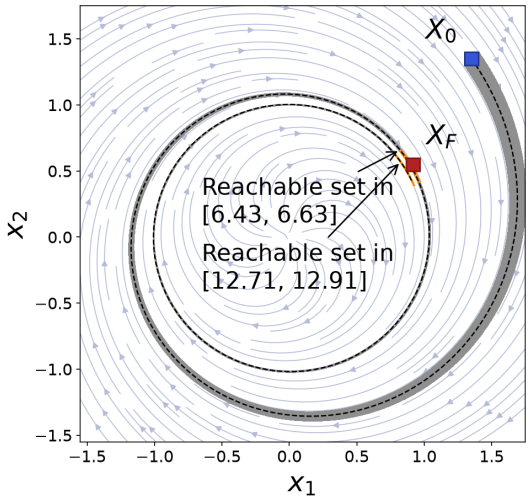


Figure 15: Reachable set of the nonlinear system with stable limit cycle from initial set X_0 for duration $[0, 6.426] \cup [6.628, 12.709]$ (gray), estimated reach-time bound $I_0 = [6.426, 6.628]$, $I_1 = [12.709, 12.911]$ (orange) with a simulated periodic reachable trajectory (black dashed lines).

2.3 Benchmark NL-SLC_UNR_BOX

Config ID: `NL_SLC_FIN_UNR_BOX`.

2.3.1 Specification

- Initial Set: $X_0 = [1.3, 1.4] \times [1.3, 1.4]$

- Target Set: $X_F = [0.475, 0.525] \times [0.475, 0.525]$
- Time Horizon: $[0, 15]$
- Expected Result: Unreachable

2.3.2 Results

KRTB		CORA	JuliaReach	dReach	hj_reachability	SOSTOOLS
Eigenpairs	Bounds					
Analytical	Unr (0.01)	S/E	Unr (2.48)	T/O	Unr (267.15)	N/A[a]

[a] Not directly applicable due to trigonometric, exponential or fractional polynomial functions.

KRTB reach-time bounds: $\emptyset$. Simulation baseline: none of 2004 sampled trajectories reaches the target; the closest approach (distance) is 0.2598.

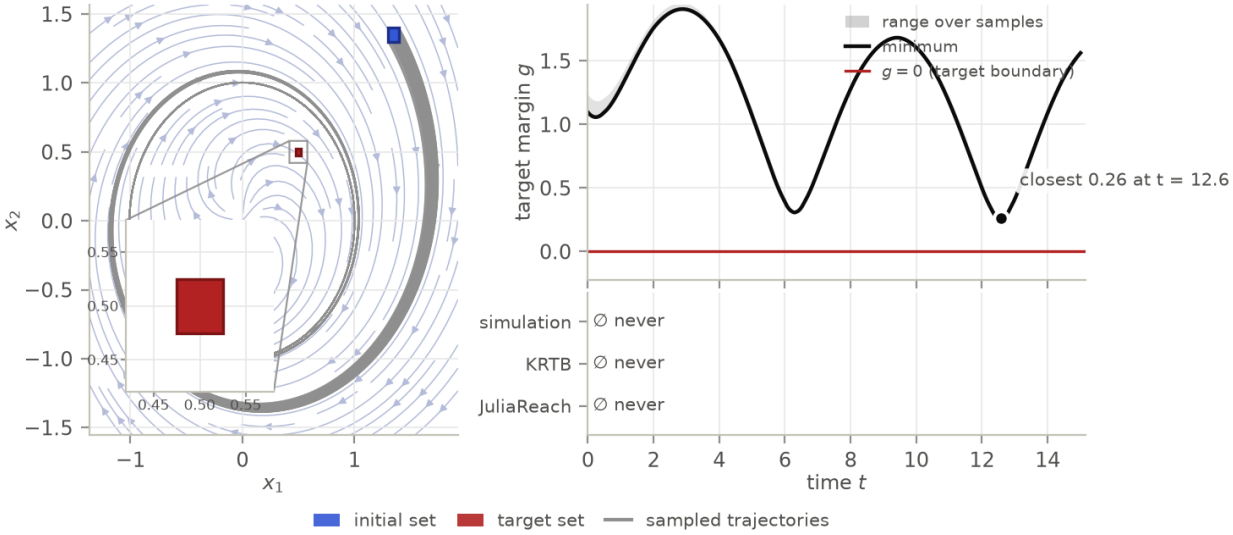
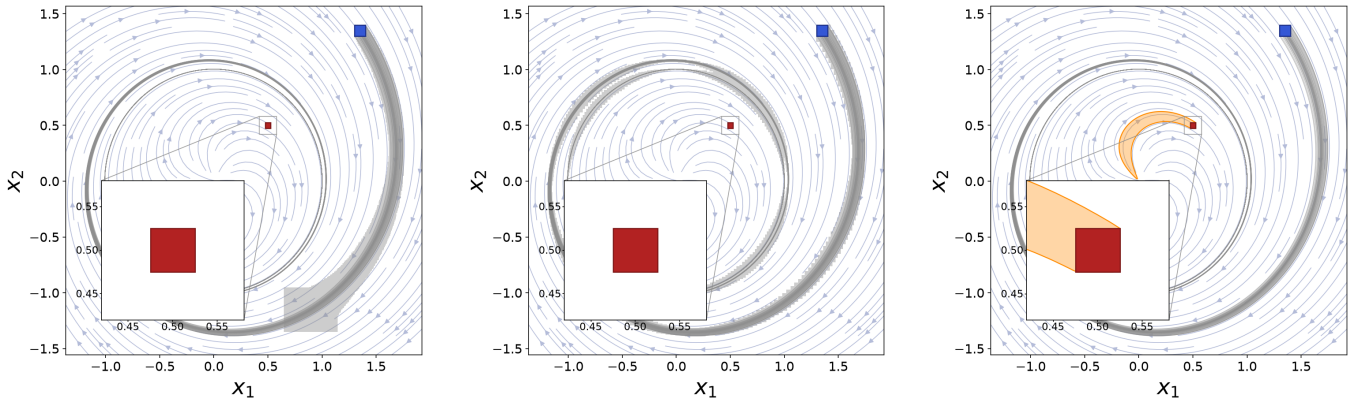


Figure 16: NL-SLC_UNR_BOX: 84 sampled trajectories from X_0 (grey), the target margin $g(x(t))$ ($g \leq 0$ inside X_F) and, on the same time axis, the simulated in-target times (none), the KRTB reach-time bounds and the times of the JuliaReach flowpipe segments that may meet the target.



(a) CORA: S/E. (b) JuliaReach: Unr (2.48). (c) hj_reachability: Unr (267.15).

Figure 17: NL-SLC_UNR_BOX: what the tools computed, drawn in the phase plane with the sampled trajectories (grey). CORA and JuliaReach: the flowpipe, with the segments disjoint from the target in grey, those that may meet it in light orange and those inside it in orange; hj_reachability: the states from which the target is reached ($V \leq 0$, light orange).

3 Nonlinear System with Known Eigenfunctions (Saddle Point System)

3.1 System

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = [\nabla\Psi(x)]^{-1}\begin{bmatrix}-1 & 0\\ 0 & 2.5\end{bmatrix}\Psi(x)$$

where the principal eigenfunctions are denoted as $\Psi(x) = [\psi_1(x), \psi_2(x)]^T$ with $\psi_1(x) = x_1^2 + 2x_2 + x_2^3$ and $\psi_2(x) = x_1 + \sin(x_2) + x_1^3$, and the associated eigenvalues are $\lambda_1 = -1$ and $\lambda_2 = 2.5$ at the unstable equilibrium $(0, 0)$.

Both benchmarks are backward reachability problems. They ask whether the trajectories of the time-reversed flow $\dot{x} = -f(x)$ from X_F reach X_0 within the horizon, which is the same as asking whether the trajectories of $\dot{x} = f(x)$ from X_0 reach X_F. The configs state them in the time-reversed form, with X_F as the initial set and X_0 as the target set. The results and figures below follow the configs. Their times are those of the time-reversed flow, and in the figures the initial set (blue) is X_F and the target set (red) is X_0.

3.2 Benchmark NL-EIG_BWD_REA_BOX

Config ID: `NL_EIG_FIN_BACKWARD_REA_BOX`.

3.2.1 Specification

- Initial Set: $X_0 = [0, 0.1] \times [1.1, 1.2]$
- Target Set: $X_F = [1.8, 1.9] \times [-0.8, -0.7]$
- Time Horizon: $[0, 1.3]$
- Expected Result: Reachable

3.2.2 Results

KRTB		CORA	JuliaReach	dReach	hj_reachability	SOSTOOLS
Eigenpairs	Bounds					
Analytical	Rea (0.01)	N/A	N/A	T/O	Rea (6.95)	N/A[a]

[a] Not directly applicable due to trigonometric, exponential or fractional polynomial functions.

KRTB reach-time bounds: $[0.760, 0.884]$. Simulation baseline: 899 of 2004 sampled trajectories reach the target, at the times $[0.797, 0.877]$. The problem is a backward one: trajectories follow $\dot{x} = -f(x)$.

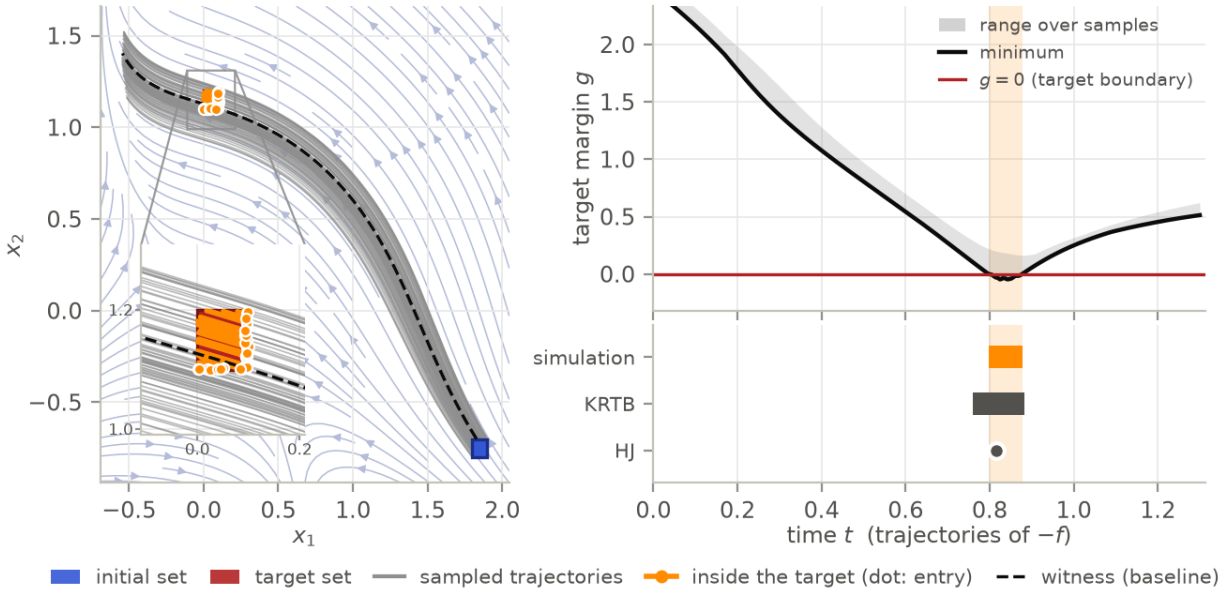
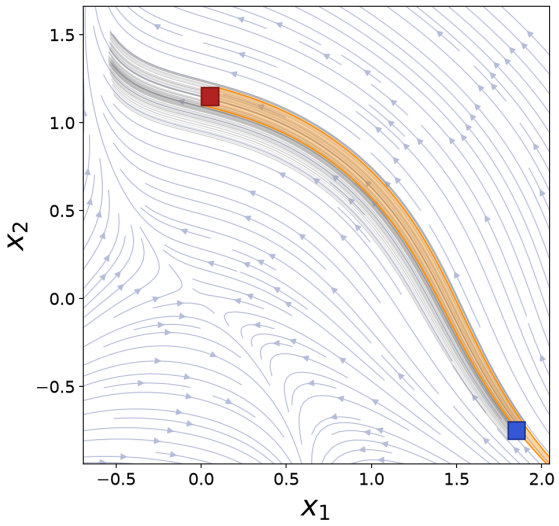


Figure 18: NL-EIG_BWD_REA_BOX: 84 sampled trajectories from X_0 (grey; orange inside X_F, with a dot at the entry), the witness of the simulation baseline (dashed), the target margin $g(x(t))$ ($g \leq 0$ inside X_F) and, on the same time axis, the simulated in-target times (orange band), the KRTB reach-time bounds and the first reach time estimated by hj_reachability (dot).



(a) hj_reachability: Rea (6.95).

Figure 19: NL-EIG_BWD_REA_BOX: what the tools computed, drawn in the phase plane with the sampled trajectories (grey). Hj_reachability: the states from which the target is reached ($V \leq 0$, light orange).

3.3 Benchmark NL-EIG_BWD_UNR_BOX

Config ID: `NL_EIG_FIN_BACKWARD_UNR_BOX`.

3.3.1 Specification

- Initial Set: $X_0 = [0, 0.1] \times [1.1, 1.2]$
- Target Set: $X_F = [1.5, 1.6] \times [0.1, 0.2]$
- Time Horizon: $[0, 1.3]$
- Expected Result: Unreachable

3.3.2 Results

KRTB		CORA	JuliaReach	dReach	hj_reachability	SOSTOOLS
Eigenpairs	Bounds					
Analytical	Unr (0.01)	N/A	N/A	T/O	Unr (7.87)	N/A[a]

[a] Not directly applicable due to trigonometric, exponential or fractional polynomial functions.

KRTB reach-time bounds: $\emptyset$. Simulation baseline: none of 2004 sampled trajectories reaches the target; the closest approach (distance) is 0.06195. The problem is a backward one: trajectories follow $\dot{x} = -f(x)$.

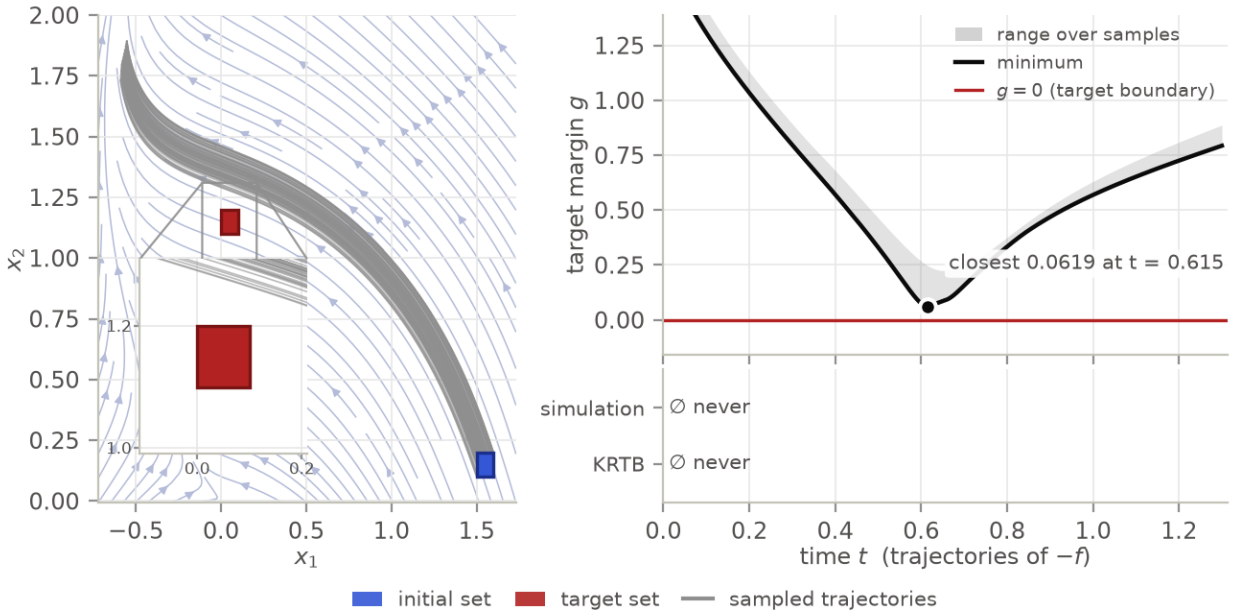
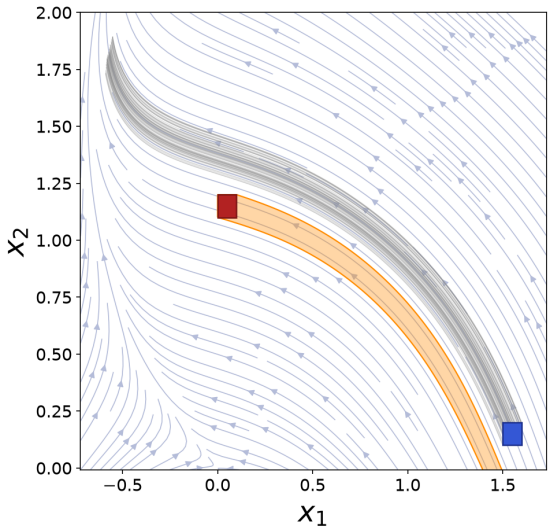


Figure 20: NL-EIG_BWD_UNR_BOX: 84 sampled trajectories from X_0 (grey), the target margin $g(x(t))$ ($g \leq 0$ inside X_F) and, on the same time axis, the simulated in-target times (none) and the KRTB reach-time bounds.



(a) hj_reachability: Unr (7.87).

Figure 21: NL-EIG_BWD_UNR_BOX: what the tools computed, drawn in the phase plane with the sampled trajectories (grey). Hj_reachability: the states from which the target is reached ($V \leq 0$, light orange).

4 Cart-pole System (Example 2)

4.1 System

$$\dot{x} = \begin{bmatrix} \dot{p} \\ \dot{v} \\ \dot{\theta} \\ \dot{\omega} \end{bmatrix} = \begin{bmatrix} v \\ \frac{[F+m_p \sin(\theta)(l\omega^2+g\cos(\theta))]}{m_c+m_p \sin(\theta)^2} \\ \omega \\ \frac{[-F\cos(\theta)-m_p l\omega^2 \cos(\theta)\sin(\theta)-(m_c+m_p)g\sin(\theta)]}{l(m_c+m_p \sin(\theta)^2)} \end{bmatrix}$$

where $m_c = 2, m_p = 1, l = 1$, $g = 9.8065$, and θ is the angle of the pole from the downward vertical (Tedrake's cart-pole as printed; the origin is the hanging equilibrium). The system is equipped with the LQR controller $F = u(x) = -2.907\,p - 4.877\,v + 3.770\,\theta + 1.640\,\omega$ (weights $Q = \mathrm{diag}(56.53, 27.43, 1.869, 77.92)$, $R = 6.688$ for the linearisation at the hanging equilibrium; closed-loop eigenvalues $-0.771 \pm 0.629j$, $-0.858 \pm 3.698j$). The unsafe sets are restricted to the state set $X = [-0.06, 4] \times [-0.88, 0.7] \times [-0.83, 0.66] \times [-1.72, 2.82]$.

4.2 Benchmark CP-LQR_REA_CVG

Config ID: `CP_LQR_FIN_REA_CONVERGE`.

4.2.1 Specification

- Initial Set: $X_0 = [0.8, 0.3, -\frac{\pi}{4}, 0.2]$
- Target Set: $X_F = \{x \mid 10^{-30} \leq \|x\|_\infty \leq 10^{-3}, x \in X\}$ (a box shell: the equilibrium $x_e = 0$ is excluded, as the time-to-reach bounds require)
- Time Horizon: $[0, 12]$
- Expected Result: Reachable

4.2.2 Results

KRTB		CORA	JuliaReach	dReach	hj_reachability	SOSTOOLS
Eigenpairs	Bounds					
6.82	Rea (0.02)	Rea (14.71)	Rea (14.85)	δ-Rea (9.46)	OOM	N/A[a]

[a] Not directly applicable due to trigonometric, exponential or fractional polynomial functions.

KRTB reach-time bounds: $[8.168, 12]$. Simulation baseline: 1 of 1 sampled trajectories reach the target, at the times $[9.952, 12]$.

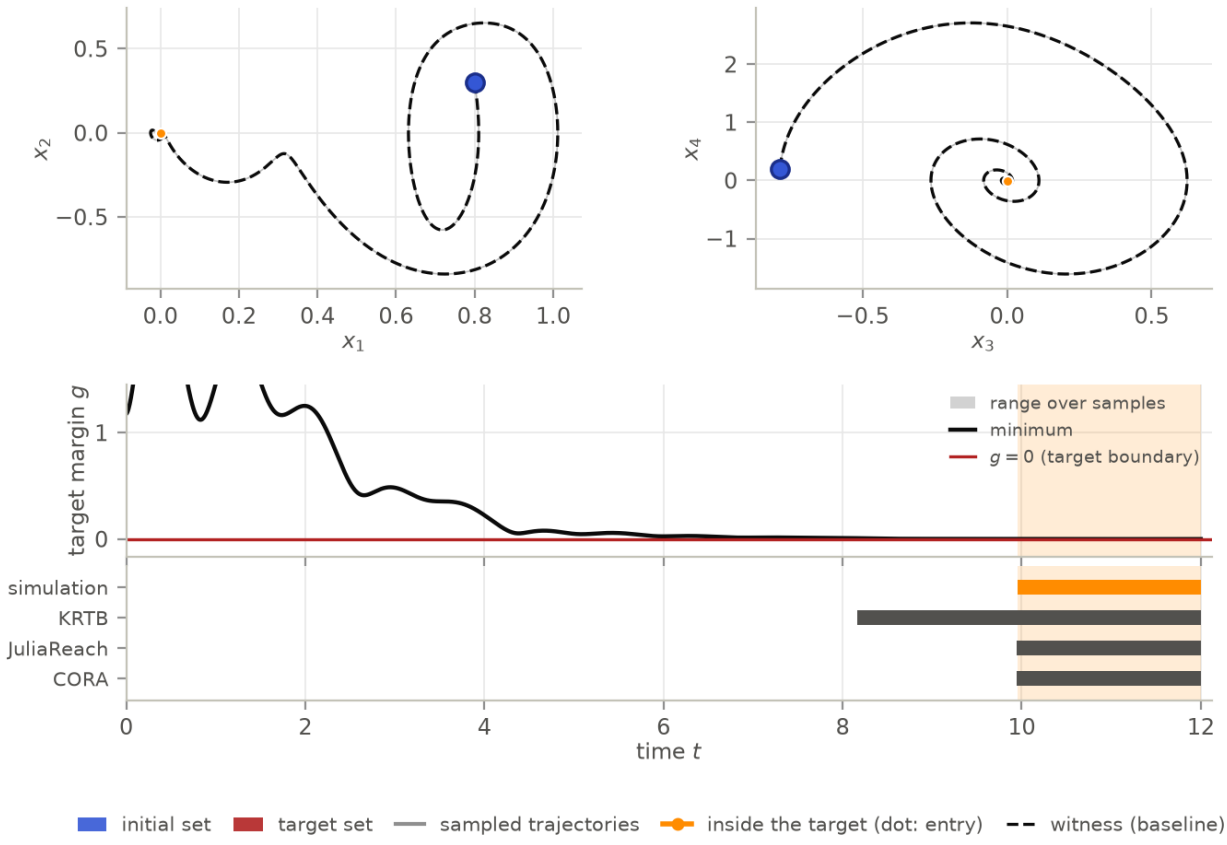
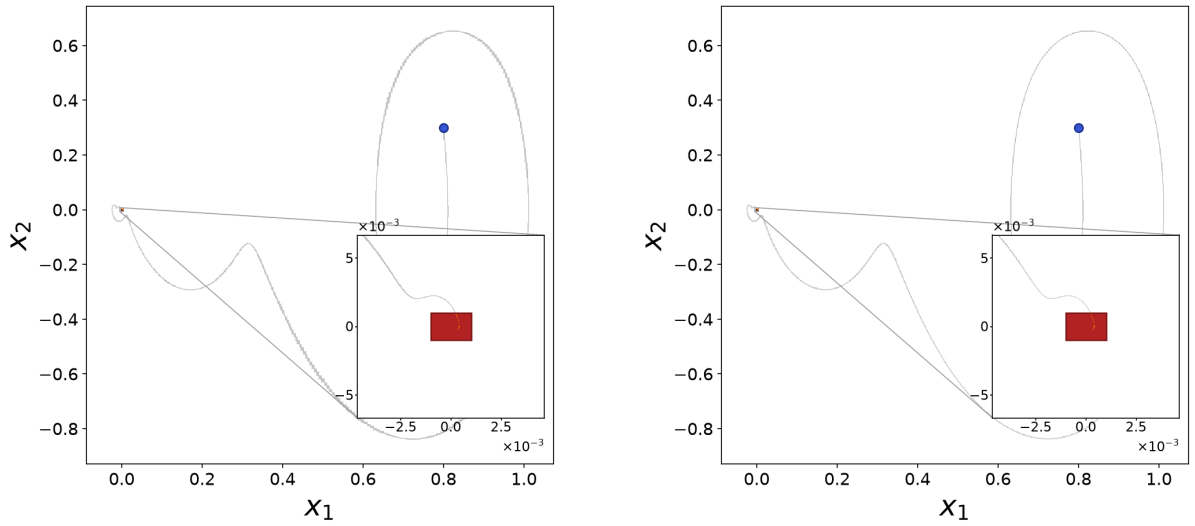


Figure 22: CP-LQR_REA_CVG: the trajectory from the initial state (grey; orange inside X_F, with a dot at the entry), the witness of the simulation baseline (dashed), the target margin $g(x(t))$ ($g \leq 0$ inside X_F) and, on the same time axis, the simulated in-target times (orange band), the KRTB reach-time bounds, the times of the JuliaReach flowpipe segments that may meet the target and the times of the CORA flowpipe segments that may meet the target.



(a) CORA: Rea (14.71).

(b) JuliaReach: Rea (14.85).

Figure 23: CP-LQR_REA_CVG: what the tools computed, drawn in the projection with the trajectory (grey). CORA and JuliaReach: the flowpipe, with the segments disjoint from the target in grey, those that may meet it in light orange and those inside it in orange.

4.3 Benchmark CP-LQR_REA_USF

Config ID: `CP_LQR_FIN_REA_UNSAFE`.

4.3.1 Specification

- Initial Set: $X_0 = [0.8, 0.3, -\frac{\pi}{4}, 0.2]$
- Target Set: $X_F = \{x \mid p + l \sin(\theta) > 1.3, x \in X\}$
- Time Horizon: $[0, 10]$
- Expected Result: Reachable

4.3.2 Results

KRTB		CORA	JuliaReach	dReach	hj_reachability	SOSTOOLS
Eigenpairs	Bounds					
55.91	Rea (0.01)	N/A	N/A	δ-Rea (2.95)	OOM	N/A[a]

[a] Not directly applicable due to trigonometric, exponential or fractional polynomial functions.

KRTB reach-time bounds: $[0, 1.723]$. Simulation baseline: 1 of 1 sampled trajectories reach the target, at the times $[0.829, 1.143]$.

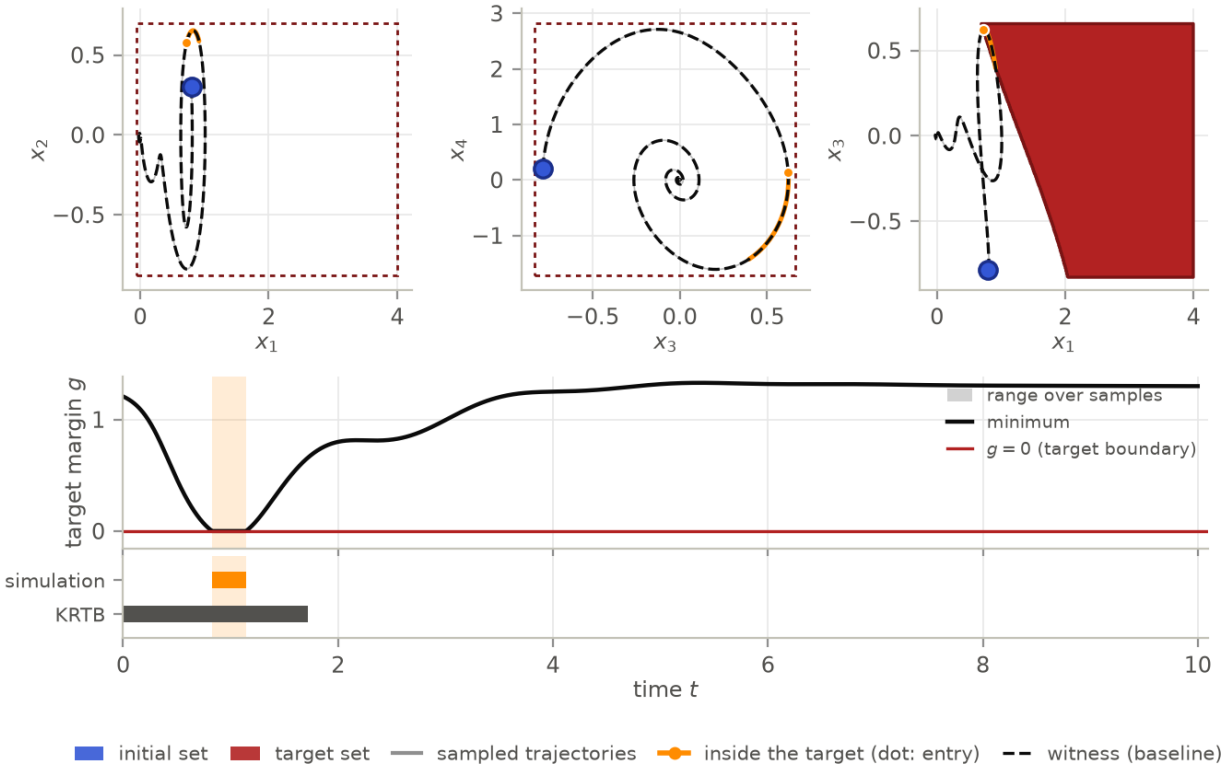


Figure 24: CP-LQR_REA_USF: the trajectory from the initial state (grey; orange inside X_F, with a dot at the entry), the witness of the simulation baseline (dashed), the target margin $g(x(t))$ ($g \leq 0$ inside X_F) and, on the same time axis, the simulated in-target times (orange band) and the KRTB reach-time bounds. Sets drawn as dotted boxes (x_1–x_2, x_3–x_4) are bounding boxes: they cannot be drawn exactly in that plane.

4.4 Benchmark CP-LQR_UNR_SAF

Config ID: `CP_LQR_FIN_UNR_SAFE`.

4.4.1 Specification

- Initial Set: $X_0 = [0.8, 0.3, -\frac{\pi}{4}, 0.2]$
- Target Set: $X_F = \{x \mid p + l \sin(\theta) > 3.7, x \in X\}$
- Time Horizon: $[0, 10]$
- Expected Result: Unreachable

4.4.2 Results

KRTB		CORA	JuliaReach	dReach	hj_reachability	SOSTOOLS
Eigenpairs	Bounds					
54.49	Unr (0.01)	N/A	N/A	Unr (0.55)	OOM	N/A[a]

[a] Not directly applicable due to trigonometric, exponential or fractional polynomial functions.

KRTB reach-time bounds: $\emptyset$. Simulation baseline: none of 1 sampled trajectories reaches the target; the closest approach (value of the target's defining function) is 2.341.

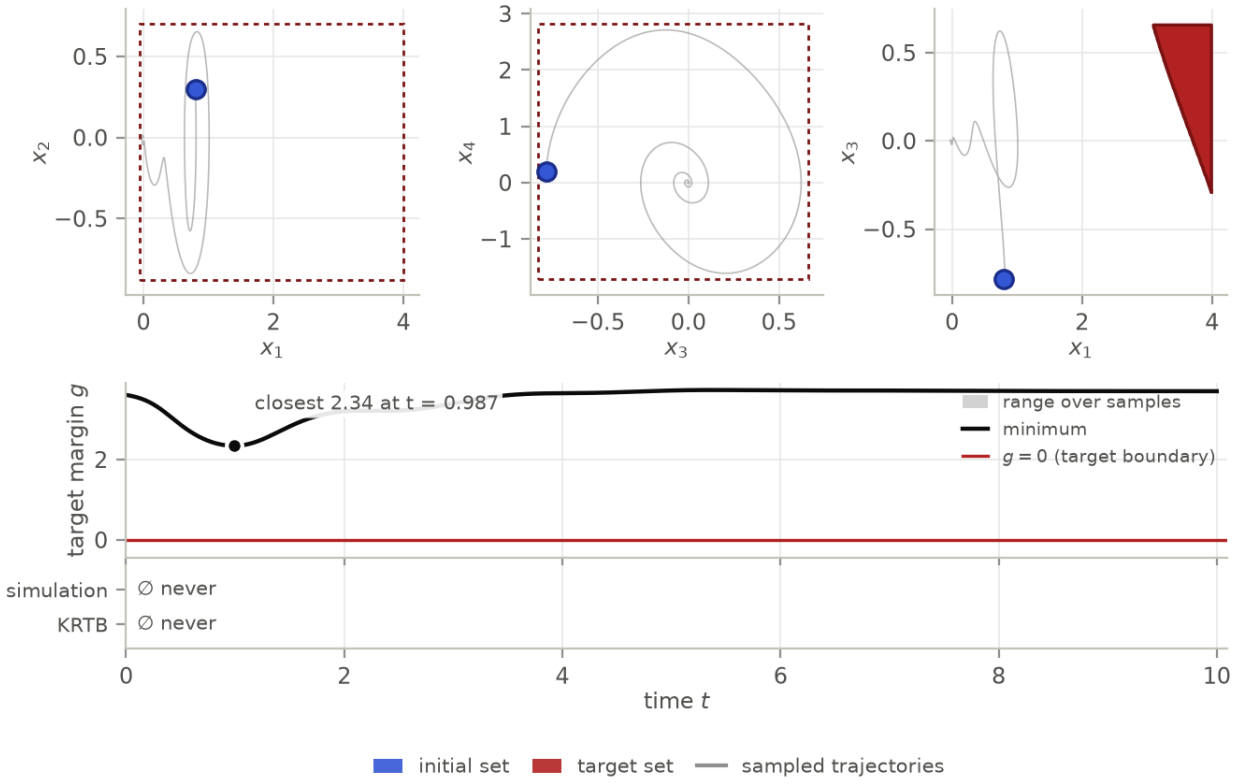


Figure 25: CP-LQR_UNR_SAF: the trajectory from the initial state (grey), the target margin $g(x(t))$ ($g \leq 0$ inside X_F) and, on the same time axis, the simulated in-target times (none) and the KRTB reach-time bounds. Sets drawn as dotted boxes (x_1–x_2, x_3–x_4) are bounding boxes: they cannot be drawn exactly in that plane.

5 Multi-agent Consensus System (Example 3)

5.1 System

As defined in Fig. 26.

In the target sets below, $X = [-3, 3]^{16}$ is the state set of the configs.

5.2 Benchmark MAS-CON_REA_BOX

Config ID: `MAS_CON_FIN_REA_BOX`.

5.2.1 Specification

- Initial Set: $X_0 = \{x \in \mathbb{R}^{16} \mid \|x - c\|_\infty \leq 0.1\}$, where

$$c^\top = [2.3757, -1.1287, -1.6545, 1.8835, 2.0459, -1.5123, -0.2924, 1.0962,$$
$$1.7267, -1.6586, 0.8248, 1.5392, 0.2486, -1.6764, -2.3224, -1.0923]$$

(rounded to four decimals; Table I of the paper gives the same sets agent by agent)

- Target Set: $X_F = \{x \in \mathbb{R}^{16} \mid 10^{-30} \leq \|x\|_\infty \leq 0.1, x \in X\}$ (a box shell, as for CP-LQR_REA_CVG)
- Time Horizon: $[0, 7]$
- Expected Result: Reachable

5.2.2 Results

KRTB		CORA	JuliaReach	dReach	hj_reachability	SOSTOOLS
Eigenpairs	Bounds					
56.93	Rea (0.01)	Rea (52.18)	Rea (47.37)	T/O	OOM	N/A[a]

[a] Not directly applicable due to trigonometric, exponential or fractional polynomial functions.

KRTB reach-time bounds: $[4.949, 7]$. Simulation baseline: 2000 of 2000 sampled trajectories reach the target, at the times $[5.735, 7]$.

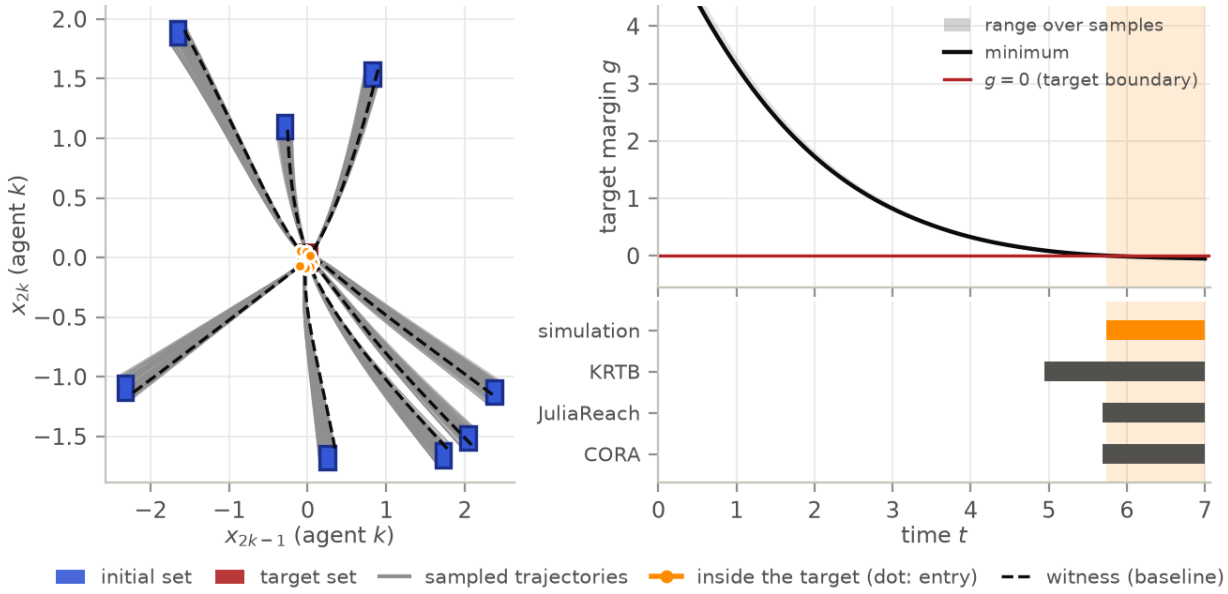
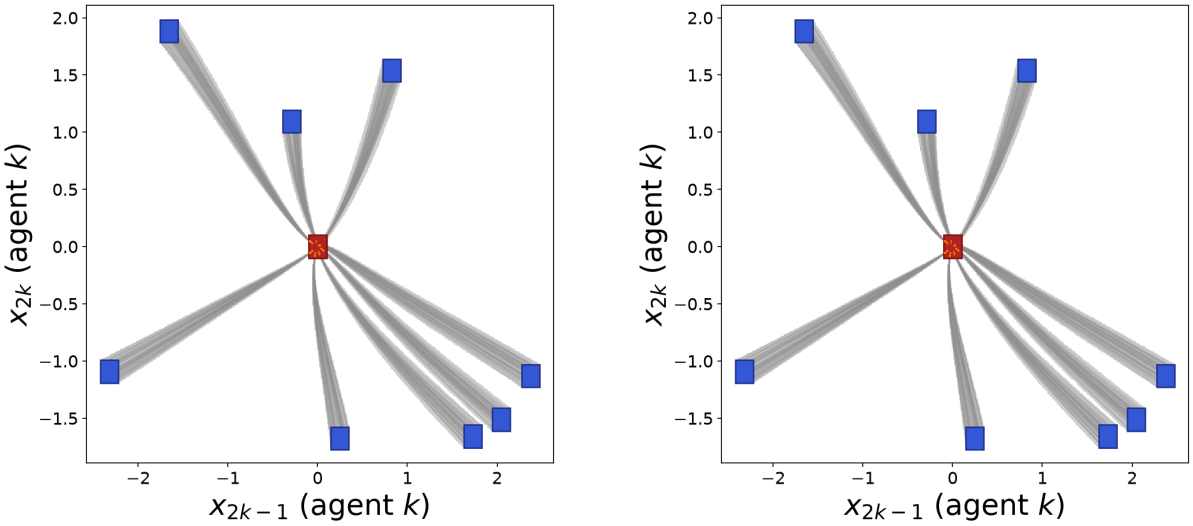


Figure 27: MAS-CON_REA_BOX: 80 sampled trajectories from X_0 (grey; orange inside X_F, with a dot at the entry), the witness of the simulation baseline (dashed), the target margin $g(x(t))$ ($g \leq 0$ inside X_F) and, on the same time axis, the simulated in-target times (orange band), the KRTB reach-time bounds, the times of the JuliaReach flowpipe segments that may meet the target and the times of the CORA flowpipe segments that may meet the target.



(a) CORA: Rea (52.18).

(b) JuliaReach: Rea (47.37).

Figure 28: MAS-CON_REA_BOX: what the tools computed, drawn in the plane of the agents with the sampled trajectories (grey). CORA and JuliaReach: the flowpipe, with the segments disjoint from the target in grey, those that may meet it in light orange and those inside it in orange.

5.3 Benchmark MAS-CON_UNR_SAF_0

Config ID: `MAS_CON_FIN_UNR_SAFE_0`.

5.3.1 Specification

- Initial Set: $X_0 = \{x \in \mathbb{R}^{16} \mid \|x - c\|_\infty \leq 0.1\}$, where

$$c^\top = [2.3757, -1.1287, -1.6545, 1.8835, 2.0459, -1.5123, -0.2924, 1.0962,$$
$$1.7267, -1.6586, 0.8248, 1.5392, 0.2486, -1.6764, -2.3224, -1.0923]$$

  (rounded to four decimals; Table I of the paper gives the same sets agent by agent)

- Target Set: $X_F = \{x \in \mathbb{R}^{16} \mid \|x - c_F\|_\infty \leq 0.25, x \in X\}$, where $c_F = [1.25, \ldots, 1.25]^\top$ (the unsafe set X_U^1 of the paper)
- Time Horizon: $[0, 7]$
- Expected Result: Unreachable

5.3.2 Results

KRTB		CORA	JuliaReach	dReach	hj_reachability	SOSTOOLS
Eigenpairs	Bounds					
60.40	Unr (0.01)	Unr (51.76)	Unr (43.81)	Unr (8.21)	OOM	N/A[a]

[a] Not directly applicable due to trigonometric, exponential or fractional polynomial functions.

KRTB reach-time bounds: $\emptyset$. Simulation baseline: none of 2000 sampled trajectories reaches the target; the closest approach (distance) is 4.029.

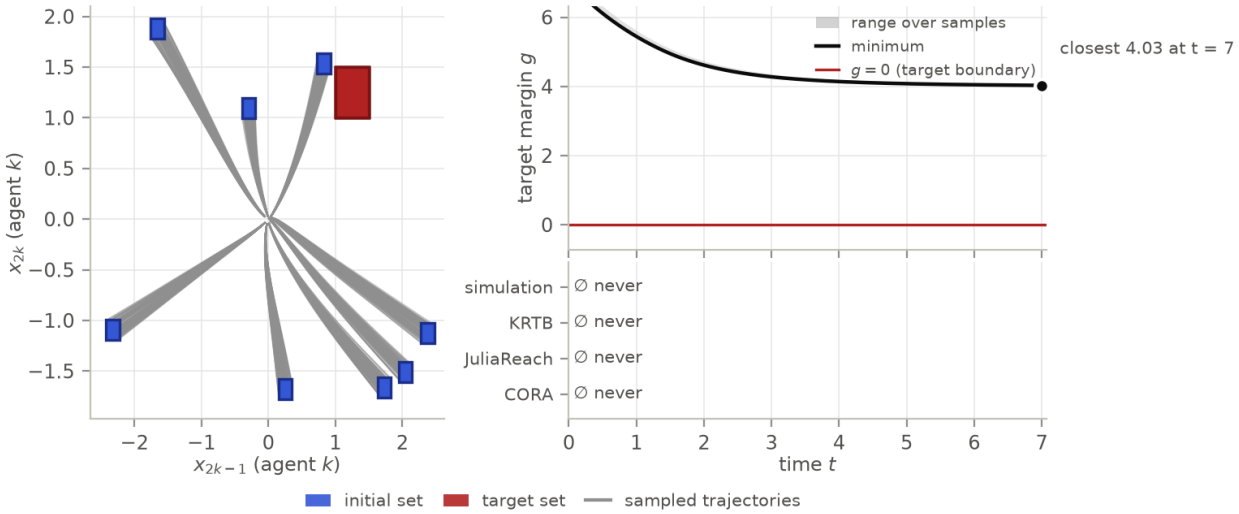
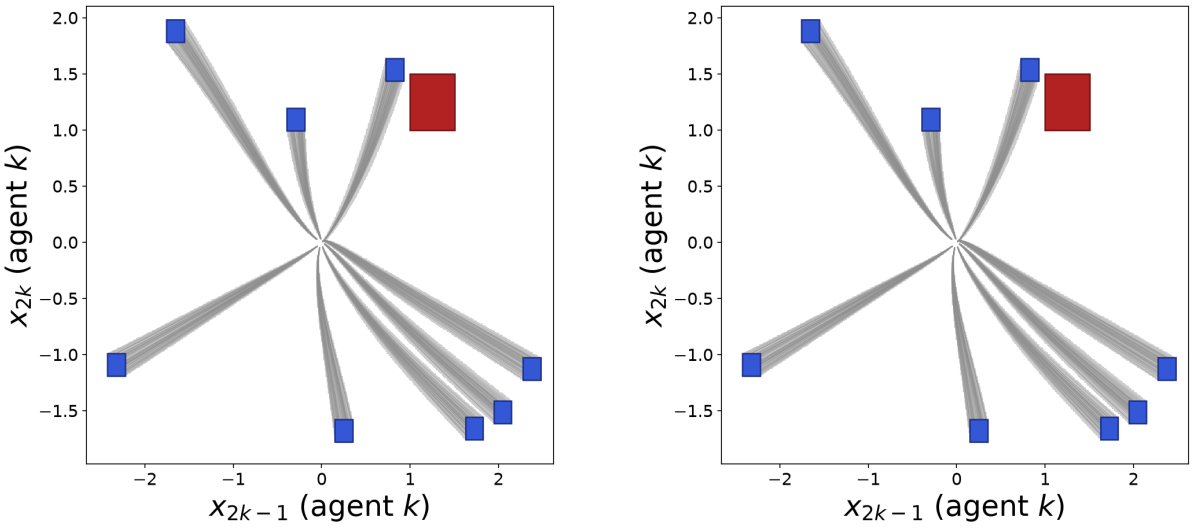


Figure 29: MAS-CON_UNR_SAF_0: 80 sampled trajectories from X_0 (grey), the target margin $g(x(t))$ ($g \leq 0$ inside X_F) and, on the same time axis, the simulated in-target times (none), the KRTB reach-time bounds, the times of the JuliaReach flowpipe segments that may meet the target and the times of the CORA flowpipe segments that may meet the target.



(a) CORA: Unr (51.76).

(b) JuliaReach: Unr (43.81).

Figure 30: MAS-CON_UNR_SAF_0: what the tools computed, drawn in the plane of the agents with the sampled trajectories (grey). CORA and JuliaReach: the flowpipe, with the segments disjoint from the target in grey, those that may meet it in light orange and those inside it in orange.

5.4 Benchmark MAS-CON_UNR_SAF_1

Config ID: `MAS_CON_FIN_UNR_SAFE_1`.

5.4.1 Specification

- Initial Set: $X_0 = \{x \in \mathbb{R}^{16} \mid \|x - c\|_\infty \leq 0.1\}$, where

$$c^\top = [2.3757, -1.1287, -1.6545, 1.8835, 2.0459, -1.5123, -0.2924, 1.0962,$$
$$1.7267, -1.6586, 0.8248, 1.5392, 0.2486, -1.6764, -2.3224, -1.0923]$$

  (rounded to four decimals; Table I of the paper gives the same sets agent by agent)

- Target Set: $X_F = \{x \in \mathbb{R}^{16} \mid \|x - c_F\|_\infty \leq 0.1, x \in X\}$, where $c_F = [-1.4, \ldots, -1.4]^\top$ (the unsafe set X_U^2 of the paper)
- Time Horizon: $[0, 7]$
- Expected Result: Unreachable

5.4.2 Results

KRTB		CORA	JuliaReach	dReach	hj_reachability	SOSTOOLS
Eigenpairs	Bounds					
60.45	Unr (0.01)	Unr (52.18)	Unr (43.32)	Unr (8.41)	OOM	N/A[a]

[a] Not directly applicable due to trigonometric, exponential or fractional polynomial functions.

KRTB reach-time bounds: ∅. Simulation baseline: none of 2000 sampled trajectories reaches the target; the closest approach (distance) is 5.087.

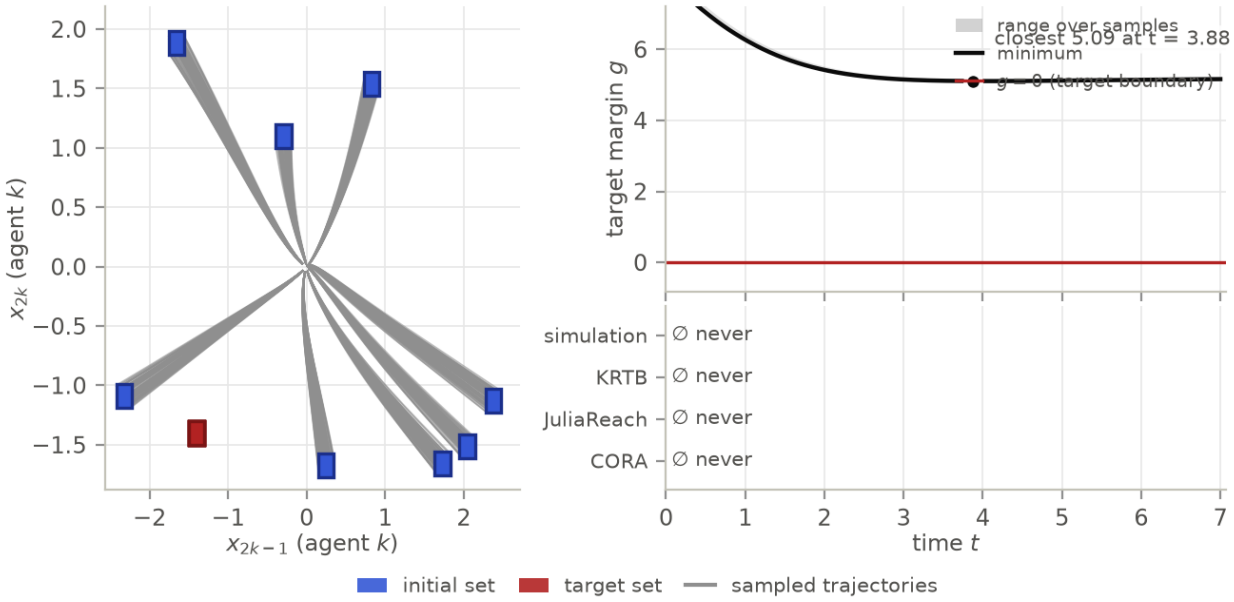
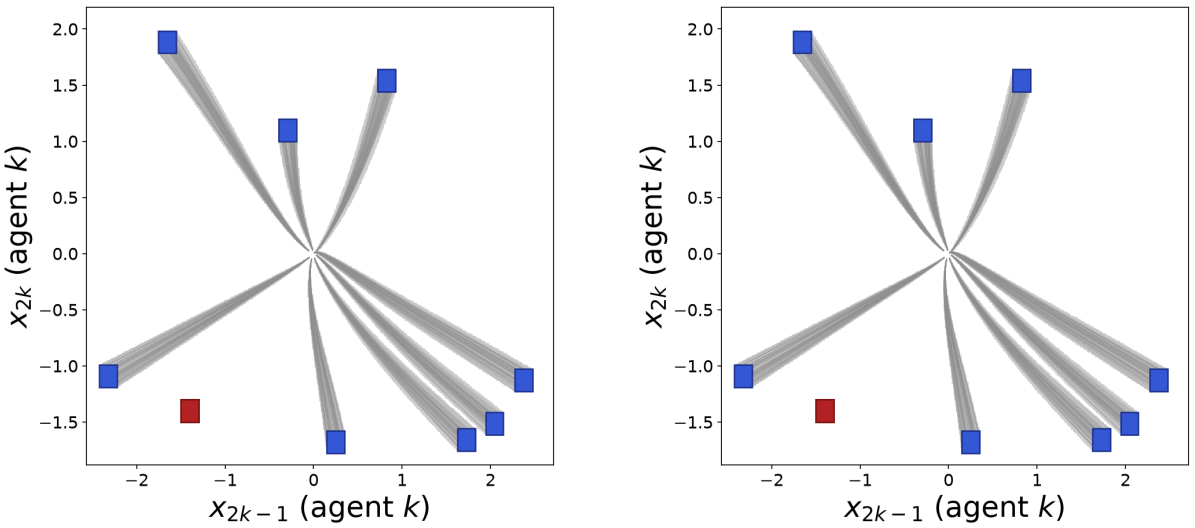


Figure 31: MAS-CON_UNR_SAF_1: 80 sampled trajectories from X_0 (grey), the target margin $g(x(t))$ ($g \leq 0$ inside X_F) and, on the same time axis, the simulated in-target times (none), the KRTB reach-time bounds, the times of the JuliaReach flowpipe segments that may meet the target and the times of the CORA flowpipe segments that may meet the target.



(a) CORA: Unr (52.18).

(b) JuliaReach: Unr (43.32).

Figure 32: MAS-CON_UNR_SAF_1: what the tools computed, drawn in the plane of the agents with the sampled trajectories (grey). CORA and JuliaReach: the flowpipe, with the segments disjoint from the target in grey, those that may meet it in light orange and those inside it in orange.

The dynamics of the i-th state of the 16-dimensional system (8 agents with planar positions; x_{2k-1} and x_{2k} are the coordinates of agent k), for $i \in \{1, \ldots, 16\}$, are governed by the general form:

$$\dot{x}_i = -\alpha x_i - \beta \cdot \frac{F_i(\mathbf{x})}{1 + e^{-x_i}} \tag{1}$$

where $\mathbf{x} = [x_1, \ldots, x_{16}]^T$ is the joint state vector, $\alpha = 0.4419$ is the uniform self-damping rate, and $\beta = 0.1$ is the coupling strength. The nonlinear coupling functions $F_i(\mathbf{x})$, which define the network's topology and interaction laws, are explicitly defined below.

$$\begin{aligned}
F_1(\mathbf{x}) = {} & 0.7799\sin(x_1 - x_2) + 0.4384\sin(x_1 - x_3) \\
& + 0.7235\sin(x_1 - x_4) + 0.5385\sin(x_1 - x_6) \\
& + 0.9359\tanh(x_1 - x_2) + 0.5261\tanh(x_1 - x_3) \\
& + 0.8682\tanh(x_1 - x_4) + 0.6462\tanh(x_1 - x_6)
\end{aligned}$$

$$\begin{aligned}
F_2(\mathbf{x}) = {} & -0.2684\sin(x_1 - x_2) + 0.8037\sin(x_2 - x_4) \\
& + 0.0659\sin(x_2 - x_6) + 0.2881\sin(x_2 - x_7) \\
& + 0.9096\sin(x_2 - x_8) - 0.3221\tanh(x_1 - x_2) \\
& + 0.9645\tanh(x_2 - x_4) + 0.0791\tanh(x_2 - x_6) \\
& + 0.3458\tanh(x_2 - x_7) + 1.0915\tanh(x_2 - x_8)
\end{aligned}$$

$$\begin{aligned}
F_3(\mathbf{x}) = {} & -0.2134\sin(x_1 - x_3) + 0.0249\sin(x_3 - x_4) \\
& + 0.2303\sin(x_3 - x_7) - 0.2561\tanh(x_1 - x_3) \\
& + 0.0299\tanh(x_3 - x_4) + 0.2764\tanh(x_3 - x_7)
\end{aligned}$$

$$\begin{aligned}
F_4(\mathbf{x}) = {} & -0.9091\sin(x_1 - x_4) - 0.1332\sin(x_2 - x_4) \\
& - 0.5234\sin(x_3 - x_4) + 0.6690\sin(x_4 - x_5) \\
& + 0.4908\sin(x_4 - x_8) - 1.0910\tanh(x_1 - x_4) \\
& - 0.1598\tanh(x_2 - x_4) - 0.6281\tanh(x_3 - x_4) \\
& + 0.8028\tanh(x_4 - x_5) + 0.5889\tanh(x_4 - x_8)
\end{aligned}$$

$$\begin{aligned}
F_5(\mathbf{x}) = {} & -0.8379\sin(x_4 - x_5) + 0.3140\sin(x_5 - x_6) \\
& + 0.5726\sin(x_5 - x_7) - 1.0055\tanh(x_4 - x_5) \\
& + 0.3768\tanh(x_5 - x_6) + 0.6872\tanh(x_5 - x_7)
\end{aligned}$$

$$\begin{aligned}
F_6(\mathbf{x}) = {} & -0.4528\sin(x_1 - x_6) - 0.3530\sin(x_2 - x_6) \\
& - 0.4591\sin(x_5 - x_6) + 0.4130\sin(x_6 - x_7) \\
& - 0.5434\tanh(x_1 - x_6) - 0.4236\tanh(x_2 - x_6) \\
& - 0.5509\tanh(x_5 - x_6) + 0.4956\tanh(x_6 - x_7)
\end{aligned}$$

$$\begin{aligned}
F_7(\mathbf{x}) = {} & -0.7411\sin(x_2 - x_7) - 0.4224\sin(x_3 - x_7) \\
& - 0.6344\sin(x_5 - x_7) - 0.5229\sin(x_6 - x_7) \\
& + 0.0014\sin(x_7 - x_8) - 0.8893\tanh(x_2 - x_7) \\
& - 0.5068\tanh(x_3 - x_7) - 0.7613\tanh(x_5 - x_7) \\
& - 0.6275\tanh(x_6 - x_7) + 0.0017\tanh(x_7 - x_8)
\end{aligned}$$

$$\begin{aligned}
F_8(\mathbf{x}) = {} & -0.7094\sin(x_2 - x_8) - 0.6962\sin(x_4 - x_8) \\
& - 0.0531\sin(x_7 - x_8) - 0.8513\tanh(x_2 - x_8) \\
& - 0.8354\tanh(x_4 - x_8) - 0.0638\tanh(x_7 - x_8)
\end{aligned}$$

$$\begin{aligned}
F_9(\mathbf{x}) = {} & -0.9650\sin(x_{11} - x_9) - 0.9450\sin(x_{12} - x_9) \\
& - 0.4723\sin(x_{14} - x_9) + 0.2351\sin(x_9 - x_{10}) \\
& - 1.1580\tanh(x_{11} - x_9) - 1.1341\tanh(x_{12} - x_9) \\
& - 0.5668\tanh(x_{14} - x_9) + 0.2821\tanh(x_9 - x_{10})
\end{aligned}$$

$$\begin{aligned}
F_{10}(\mathbf{x}) = {} & -0.4858\sin(x_{12} - x_{10}) - 0.3435\sin(x_{14} - x_{10}) \\
& - 0.3244\sin(x_{15} - x_{10}) - 0.3004\sin(x_{16} - x_{10}) \\
& - 0.3087\sin(x_9 - x_{10}) - 0.5830\tanh(x_{12} - x_{10}) \\
& - 0.4122\tanh(x_{14} - x_{10}) - 0.3893\tanh(x_{15} - x_{10}) \\
& - 0.3605\tanh(x_{16} - x_{10}) - 0.3705\tanh(x_9 - x_{10})
\end{aligned}$$

$$\begin{aligned}
F_{11}(\mathbf{x}) = {} & 0.7749\sin(x_{11} - x_{12}) + 0.4606\sin(x_{11} - x_{15}) \\
& + 0.1655\sin(x_{11} - x_9) + 0.9299\tanh(x_{11} - x_{12}) \\
& + 0.5528\tanh(x_{11} - x_{15}) + 0.1986\tanh(x_{11} - x_9)
\end{aligned}$$

$$\begin{aligned}
F_{12}(\mathbf{x}) = {} & -0.8005\sin(x_{11} - x_{12}) + 0.0407\sin(x_{12} - x_{13}) \\
& + 0.4758\sin(x_{12} - x_{16}) + 0.8873\sin(x_{12} - x_9) \\
& + 0.6749\sin(x_{12} - x_{10}) - 0.9606\tanh(x_{11} - x_{12}) \\
& + 0.0488\tanh(x_{12} - x_{13}) + 0.5709\tanh(x_{12} - x_{16}) \\
& + 1.0647\tanh(x_{12} - x_9) + 0.8099\tanh(x_{12} - x_{10})
\end{aligned}$$

$$\begin{aligned}
F_{13}(\mathbf{x}) = {} & -0.6587\sin(x_{12} - x_{13}) + 0.3571\sin(x_{13} - x_{14}) \\
& + 0.8128\sin(x_{13} - x_{15}) - 0.7905\tanh(x_{12} - x_{13}) \\
& + 0.4285\tanh(x_{13} - x_{14}) + 0.9754\tanh(x_{13} - x_{15})
\end{aligned}$$

$$\begin{aligned}
F_{14}(\mathbf{x}) = {} & -0.0071\sin(x_{13} - x_{14}) + 0.4631\sin(x_{14} - x_{15}) \\
& + 0.5999\sin(x_{14} - x_9) + 0.7282\sin(x_{14} - x_{10}) \\
& - 0.0086\tanh(x_{13} - x_{14}) + 0.5558\tanh(x_{14} - x_{15}) \\
& + 0.7198\tanh(x_{14} - x_9) + 0.8738\tanh(x_{14} - x_{10})
\end{aligned}$$

$$\begin{aligned}
F_{15}(\mathbf{x}) = {} & -0.8285\sin(x_{11} - x_{15}) - 0.1281\sin(x_{13} - x_{15}) \\
& - 0.2304\sin(x_{14} - x_{15}) + 0.1325\sin(x_{15} - x_{16}) \\
& + 0.6078\sin(x_{15} - x_{10}) - 0.9941\tanh(x_{11} - x_{15}) \\
& - 0.1538\tanh(x_{13} - x_{15}) - 0.2765\tanh(x_{14} - x_{15}) \\
& + 0.1590\tanh(x_{15} - x_{16}) + 0.7293\tanh(x_{15} - x_{10})
\end{aligned}$$

$$\begin{aligned}
F_{16}(\mathbf{x}) = {} & -0.7822\sin(x_{12} - x_{16}) - 0.5326\sin(x_{15} - x_{16}) \\
& + 0.5749\sin(x_{16} - x_{10}) - 0.9387\tanh(x_{12} - x_{16}) \\
& - 0.6392\tanh(x_{15} - x_{16}) + 0.6900\tanh(x_{16} - x_{10})
\end{aligned}$$

Figure 26: The complete definition of the coupling functions $F_i(\mathbf{x})$ for the 16-agent system, as referenced in Eq. (1). The parameters are provided with full precision (rounded to four decimal places for display purposes).